

CS372: Computer Graphics

Lecture Notes

Dr. Suman Kumar Maji

Assistant Professor
Department of Computer Science & Engineering
Indian Institute of Technology Patna

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A SIMPLE SCENE

Consists of simple geometrical objects

Sun - Circle

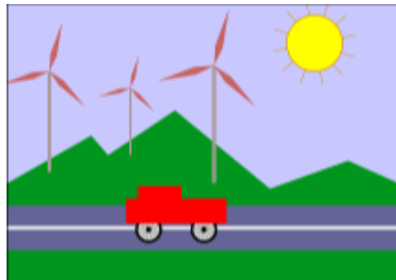
Hills - Triangle

Windmills - Polygons

Red Cart - Rectangle

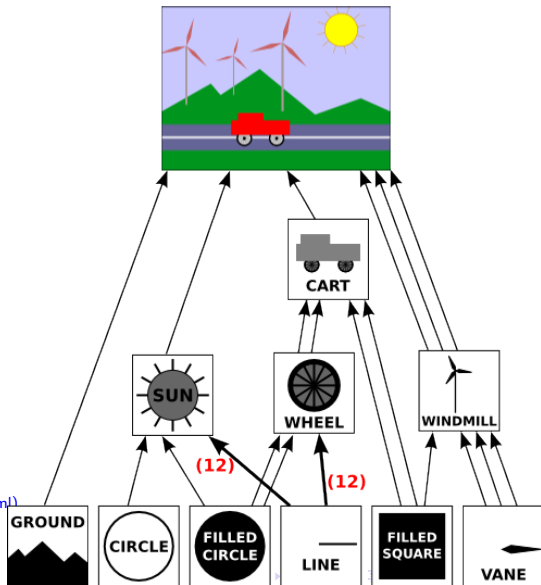
Wheels - Circle

Roads - Straight lines



UNDERSTANDING SCENE CREATION

1. Take geometrical shapes
2. Transform and create desired shapes
3. Apply color or shades
4. Incorporate



Image

courtesy

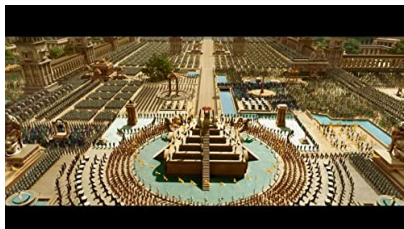
(<https://math.hws.edu/graphicsbook/c2/s4.html>)

CLASSROOM SCENE

Objects are chairs, tables, blackboards, window panes, clocks, lights, etc



MORE REALISTIC SCENES



INFERENCE

- A scene is constructed using various models (model 1, model 2, ..., model n).
- After applying a specific transformation to the models.
- Known as Modelling Transformation or Geometric Transformation in Computer Vision.

Objective

- Our objective is to understand Modelling Transformation.
- Two types: 2D and 3D transformation.

2D TRANSFORMATIONS & MATRICES

- Transformation in 2D is basically matrix transformation.
- With transformation, we can move a point, change shape, etc.
- By application of basic matrix manipulation techniques.

$$[B] = [T] [A]$$

$[A]$ = co-ordinate of points on which we apply transformation
 $[B]$ = co-ordinate of transformed points
 $[T]$ = geometric transformation matrix/operator

So, if $[A]$ and $[T]$ are known, transformed points are obtained by calculation of $[B]$.

GENERAL TRANSFORMATION OF 2D POINTS

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}; \quad \begin{aligned} x' &= ax + cy \\ y' &= bx + dy \end{aligned}$$

- $[T]$ is the transformation matrix with four scalar parameters. (x, y) are the points that are to be transformed and (x', y') are the transformed co-ordinates of (x, y) .
- So, we are pre-multiplying operator $[T]$ with $[A]$.
We can also do post-multiplication i.e. $[B] = [A] [T]$ but we have to keep the solution intact i.e. $x' = ax + cy; y' = bx + dy$.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}^T \begin{bmatrix} a & c \\ b & d \end{bmatrix}; \quad \begin{aligned} x' &= ax + cy \\ y' &= bx + dy \end{aligned}$$

SOLID BODY TRANSFORMATION

- Transformation equation is valid for all set of points and lines of the object being transformed.
- A solid transformation preserves distances between every pair of points.

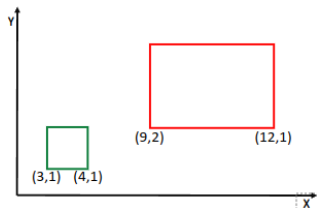
SPECIAL CASES OF 2D MATRIX

- When $a = d = 1, b = c = 0$, so $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.
 $\therefore [T] = \text{Identity matrix and } x' = x; y' = y.$
- So, when $[T] = \text{Identity}$, transformation do not change the structure of the solid body.

SCALING

- $a = d \neq 0$, so $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \implies x' = ax; y' = dy$.
- So, x is now scaled by a factor 'a' and y by a factor 'd'.

- $a = d > 1$, ENLARGEMENT
 $0 < a = d < 1$, COMPRESSION
- If a and d are same, we will have UNIFORM SCALING.
- If $a \neq d$, NON-UNIFORM SCALING.



- EXAMPLE: $a = 3, d = 2$, Non-uniform scaling $a \neq d$, Expansion $a, d > 0$
- Applicable on all points and lines of the object.

REFLECTION

When a **and/or** $d < 0$, reflection along an axis or plane (3D).

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}; x' = -x$$

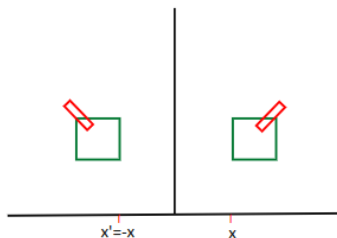
Reflection around X-axis.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}; y' = -y$$

Reflection around Y-axis.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}; \text{Special Case}$$

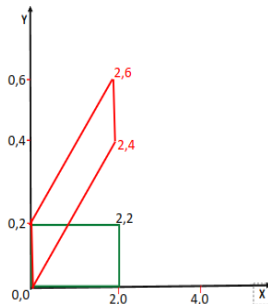
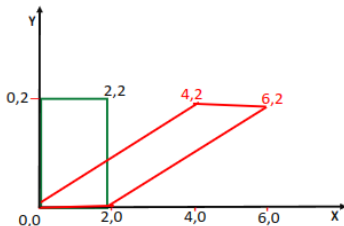
Reflection around a plane (3D case).



SHEAR

■ $a = d = 1$. Let $c = 0, b = 2$.
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

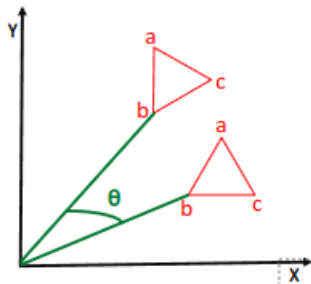
- $x' = x; y' = 2x + y$
So, y' depends linearly on x .
This effect is called Shear.
- If $c = 2, b = 0$, shear will be proportional to X-axis.



ROTATION

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- Rotation always around origin.
- Counter-clockwise direction is positive.
- $|T| = 1$ and $[T]^T = [T]^{-1}$.
- Rotation matrices are always orthogonal.



TRANSLATION

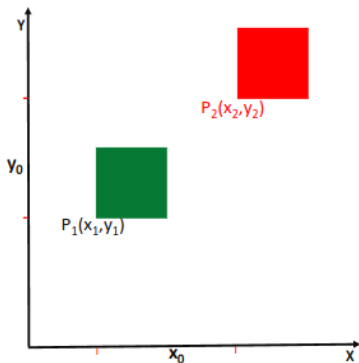
- Translate a point $P_1(x_1, y_1)$ to another point $P_2(x_2, y_2)$.
- Translation by x_0 in X and y_0 in Y, i.e.,

$$x_2 = x_0 + x_1, y_2 = y_0 + y_1.$$

- General 2D matrix equation:

$$\begin{bmatrix} B \end{bmatrix} = \begin{bmatrix} T \end{bmatrix} \begin{bmatrix} A \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$



- Translations are used in Scaling and Rotation, when objects/lines are not centered around origin.

HOMOGENEOUS COORDINATES

- In this co-ordinate system, a 2D point (x, y) is represented using a triplet (x, y, w) and the 2D transformation matrix $[T]$ will be a 3×3 matrix.

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & c & x_0 \\ b & d & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

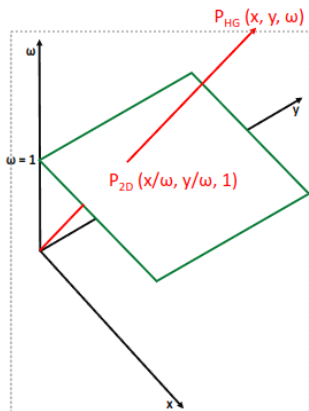
- So, (x, y, w) are co-ordinates before transformation and after transformation we get (x', y', w') .

$$\begin{cases} x' = ax + cy + wx_0 \\ y' = bx + dy + wy_0 \\ w' = w \end{cases}$$

- Remember, we are not in 3D space. We are still talking about 2D transformations. What is the role of w , also called homogeneous term?

HG CONTD...

- (x, y, w) is a point in the HG co-ordinate. Divide the first 2 elements by w i.e., $\{\frac{x}{w}, \frac{y}{w}\}$ gives the cartesian co-ordinates for the homogeneous points.
- So, (x, y, w) and (x', y', w') are the Homogeneous Representation. To get the Cartesian co-ordinates back, we divide the first 2 elements by the third.



PROPERTY

- 2 Homogeneous co-ordinates (x_1, y_1, w_1) and (x_2, y_2, w_2) may represent same point iff they are multiples of one another. Ex: $(1, 2, 3)$ & $(3, 6, 9)$.
- Hence, there is no unique homogeneous representation of a point.
- All triplets of the form $\{tx, ty, tw\}$ form a line in the x, y, w space.
- Cartesian co-ordinates are just the plane $w = 1$ in this space.
- $w = 0$? Points at infinity. $(x, y, 0)$ is the "Ideal Point".

TRANSLATION

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad w = 1$$

- Homogeneous Co-ordinate concept is created to capture translation as matrix multiplication.

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{s} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \implies \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} x \\ y \\ \frac{1}{s} \end{bmatrix} \xrightarrow[\text{Transform}]{\text{Cartesian}} \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} sx \\ sy \\ 1 \end{bmatrix}$$

- Uniform Scaling can be captured using a single parameter.

2D TRANSFORMATION MATRIX

$$T = \begin{bmatrix} a & c & p \\ b & d & q \\ m & n & s \end{bmatrix}$$

- Parameters involved in scaling, rotation, reflection, & shear:
a, b, c, d.
 - If $B = TA$, translation parameters: *p, q*.
 - If $B = AT$, translation parameters: *m, n*.
 - *s*: Special case for uniform scaling.
- If $B = TA$, what is the role of *m, n*? **Perspective Transform.**

ADVANTAGES OF HG COORDINATE

- Translation can be captured in the usual matrix multiplication framework.
- Uniform scaling can be captured by a single parameter.
- Transformations can be combined.
- Points at infinity can be captured.
- Perspective transforms can be captured.

COMPOSITE TRANSFORMATIONS

- If we want to apply a series of transformations T_1, T_2, T_3 to a point ' p ', we can do it in 2 ways.
 - 1 $p' = T_1 * p \rightarrow p'' = T_2 * p' \rightarrow p''' = T_3 * p''$
 - 2 Calculate $T = T_3 * T_2 * T_1$, and then $p''' = T * p$
- Method 2 saves large no. of computational time.

Note:

$T_3 * T_2 * T_1$. As per convention, since T_1 is applied first it has to be the rightmost transformation, and then T_2 and so on.

SOME EXAMPLES

- Translate by tx_1, ty_1 and then by tx_2, ty_2

$$\begin{bmatrix} 1 & 0 & tx_1 + tx_2 \\ 0 & 1 & ty_1 + ty_2 \\ 0 & 0 & 1 \end{bmatrix}$$

- Scale by a_1, b_1 and then by a_2, b_2

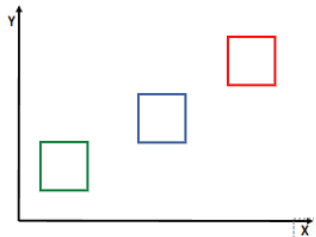
$$\begin{bmatrix} a_1 + a_2 & 0 & 0 \\ 0 & b_1 + b_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Rotate by θ_1 and then by θ_2
 - Replace θ by $(\theta_1 + \theta_2)$
 - Calculate T_1 for θ_1 and T_2 for θ_2 and multiply them

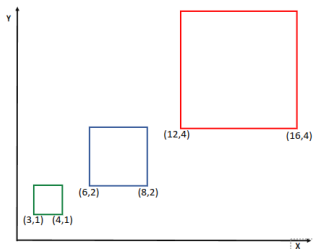
$$\begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

SOME EXAMPLES (Contd...)

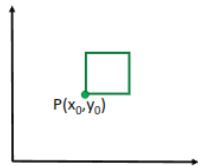
- Translate by tx_1, ty_1 and then by tx_2, ty_2



- Scale by a_1, b_1 and then by a_2, b_2



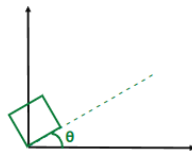
ROTATION ABOUT AN ARBITRARY POINT P



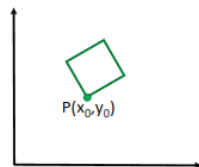
House at
 $P(x_0, y_0)$



Translate P to
origin: T_1



Rotate by θ :
 T_2



Translate back
to P: T_3

What will be the transformation matrix? T

$$T = \underbrace{\begin{bmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_3} * \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_2} * \underbrace{\begin{bmatrix} 1 & 0 & -x_0 \\ 0 & 1 & -y_0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_1}$$

SCALING ABOUT AN ARBITRARY POINT

- T_1 : Translate P to origin
- T_2 : Scale
- T_3 : Translate P back

- $$T = \underbrace{\begin{bmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_3} * \underbrace{\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_2} * \underbrace{\begin{bmatrix} 1 & 0 & -x_0 \\ 0 & 1 & -y_0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_1}$$

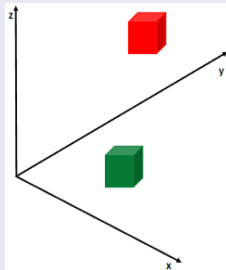
- $$T = T_3(x_0, y_0) * T_2(S_x, S_y) * T_1(-x_0, -y_0)$$

TRANSFORMATIONS IN 3D

- Homogeneous representation of a 3D point (x,y,z) is (xh,yh,zh,h) , where h is the homogeneous term.

TRANSLATION

$$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

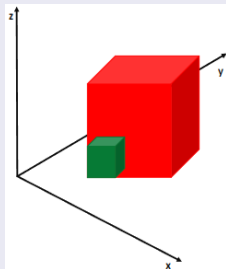


TRANSFORMATIONS IN 3D CONTD...

SCALING

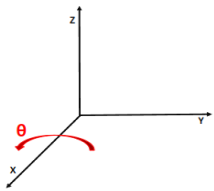
$$\begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

can be changed for
uniform scaling

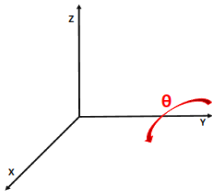


ROTATION

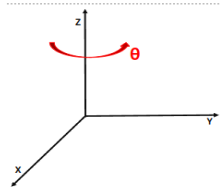
- In 2D, we were rotating around origin. In 3D, rotation will be around an axis.



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

REFLECTION

Reflection around
XY plane, Z -ve

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Reflection around
XZ plane, Y -ve

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Reflection around
YZ plane, X -ve

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Around X-axis

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Around Y-axis

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

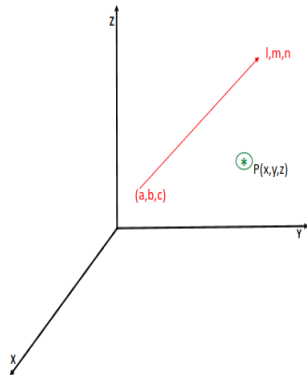
Around Z-axis

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Around origin? All diagonal elements, except 'h', -1.

ROTATION OF AN ARBITRARY POINT AROUND AN INCLINED AXIS

- 1 Translate origin to a,b,c : T_1
- 2 Rotate s.t. vector (l,m,n) coincides with one axis, say Z .
 - So, rotate s.t. vector coincides with XZ plane : T_2
 - Then rotate so that it coincides with Z -axis: T_3
- 3 Then rotate the point by θ : T_4
- 4 Reverse of 2
- 5 Reverse of 1



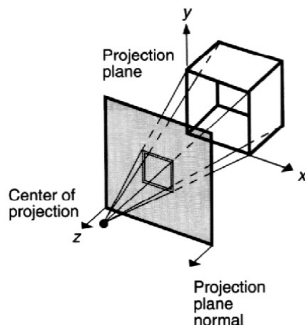
$$\therefore T = T_1^{-1} T_2^{-1} T_3^{-1} T_4 T_3 T_2 T_1$$

Projection

- Projection of a 3D object can be defined by straight **projection rays (projectors)** emanating from the object, passing/intersecting through the **projection plane** and meeting/converging to the **center of projection (COP)**.
- Broadly classified into 2 types:
 - 1 Parallel Projection
 - 1 Orthographic
 - 2 Oblique
 - 2 Perspective Projection
- COP is a finite point in 3D space.

Perspective Projection

- Distance between the COP and projection plane is finite.
- Projectors are not parallel.



- Perspective foreshortening:
 - Closer is the object to the COP, larger is its projection.
 - Further the object from the COP, smaller is its projection.
- Vanishing Points - Two parallel railway tracks appear to meet at a point on the horizon. This point is called the VP.

Vanishing Points

